ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.: | | | | | | | | | | | | | |

Course Name: B.tech **Max. Marks: 15**
Subject Name: DS **Time: 75 Minutes**
Subject Code: 25CS101 **Semester: I**

Note:
1. **Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM**
2. **Don't Write Anything on Question Paper**
3. **Attempt All Questions**
4. **Assume Any Missing Data if required**

SECTION-A *(Maximum Knowledge Level: K2)*				
Q. No.	**Question**	**CO**	**KL**	**PI**
1. Attempt All Questions				**(1 x 5)**
(a)	**Draw Memory hierarchy**	**CO1**	**K2**	**2.1.1**
(b)	**Example of syntax and semantic error**	**CO1**	**K2**	**2.1.1**
(c)	**Define the characteristics of an Algorithm**	**CO1**	**K2**	**2.1.2**
(d)	**Find the value of z=!(10*0) in the expression**	**CO2**	**K2**	**2.1.1**
(e)	**Find the value of z=++x + ++x ,where x=10**	**CO2**	**K2**	**2.1.1**
SECTION-B *(Minimum Knowledge Level: K2)*				
2. Attempt Any two Question			**(Same K Levels) (2 x 2)**	
(a)	**Difference between compiler and interpreter**	**CO1**	**K1**	**2.1.2**
(b)	**Draw flow chart for temperature conversion from Fahrenheit to Celsius(hint:-f-32=9c/5)**	**CO1**	**K1**	**2.1.1**
(c)	**Difference between Multiprogramming and Multitasking**	**CO1**	**K1**	**1.4.1**
SECTION-C *(Minimum Knowledge Level: K3)*				
3. Attempt Any one Question			**(Same K Levels) (3 x 1)**	
(a)	**Draw block Diagram of computer and explain the components**	**CO1**	**K2**	**1.4.1**
(b)	**Why Operating System also known as resource manager?**	**CO1**	**K2**	**1.4.1**
4. Attempt Any one Question			**(Same K Levels) (3 x 1)**	
(a)	**Write a C++ program to input a number and check whether it is** even or odd **using ternary operator.**	**CO2**	**K3**	**1.4.1**
(b)	**Write a C++ program to input the marks of three subjects and calculate the average marks. Display the result.**	**CO2**	**K3**	**1.4.1**

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create